Shuangning Yang

github.com/VodkaVortex | vodkavortex.github.io | realyangshuangning@gmail.com | +86 133-0866-2608

Research Interests

I am interested in both securing AI systems and using AI to improve software security. Specifically, my research focuses on improving the reliability and robustness of LLM-based systems, while exploring how LLMs can support security tasks, such as vulnerability detection.

Education

Anhui University, M.Eng. in Artificial Intelligence — Sep. 2024 – May. 2027

- Average Scores: 92/100
- Selected Papers: DetFuzz (ASE 2026 Scores: 3/3/2), IoTBec (NDSS 26, Accepted)

Hefei University of Technology, B.S. in Logistics Management — Sep. 2020 – May. 2024

- Average Scores: 88/100
- Selected Coursework: C, Java, Computer Networks, Database Principles and Applications

Publications

- **S. Yang**, H. Yang, J. Peng et al., "DetFuzz: A Semantic-Guided Fuzzing Framework via Operation Logic Inference for Black-box IoT Devices," *Manuscript*, 2026. **(ASE 2026 Scores: 3/3/2)**
- **S. Yang**, H. Yang, G. Zhao et al., "What is the Risk? Evaluating the Impact of Knowledge Distillation on LLM Vulnerabilities," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2026. **(CCF-B, Accepted)**
- H. Yang, J. Guo, **S. Yang** et al., "IoTBec: An Accurate and Efficient Recurring Vulnerability Detection Framework for Black Box IoT Devices," in *Proc. Network and Distributed System Security (NDSS)*, 2026. **(CCF-A, Accepted)**
- J. Tai, **S. Yang**, J. Wang et al., "Survey of Adversarial Attacks and Defenses for Large Language Models," in *Journal of Computer Research and Development*, 2025. **(CCF-T1 Chinese Journal, Accepted)**
- Q. Liu, H. Yang, **S. Yang** et al., "FirmSV: Detecting Stored Vulnerabilities in IoT Firmware using Static Taint Analysis," in *Proc. IEEE Int. Conf. Computer Supported Cooperative Work in Design (CSCWD)*, 2026. **(CCF-C, Accepted)**

Research Experience

DetFuzz: LLM-Driven Semantic Fuzzing for Black-box IoT Vulnerability Detection — Oct.2025 – Feb.2026

Lead Researcher

- Designed an LLM-driven fuzzing pipeline that combines CoT-guided operation logic inference with front-end static analysis to automatically identify high-risk parameters and satisfy operation prerequisites, enabling valid test case generation against black-box IoT devices without firmware or source code.
- Detected **137 vulnerabilities** (122 CVE IDs assigned; **40 previously unknown**, avg. CVSS **8.54**); outperformed SOTA black-box and grey-box fuzzers by **7**×, achieving 100% precision and 94.48% recall.

What is the Risk: Security Risks of Knowledge Distillation in LLMs — Apr.2025 – Sep.2025

Lead Researcher

- Designed an ablation-based security evaluation framework for knowledge distillation. It decouples the distillation pipeline, covering five main configurations (SeqKD, PPO/DPO-based RL/RO, and their combinations) with dual-judge assessment, and introduces a harm-level test dataset with 10 malicious categories.
- Demonstrated that joint configurations increase ASR by up to 18.38% and induces alignment failure in safety-robust models.

Engineering Experience

LLM Agent Framework for Vulnerability Detection and Exploitation — Apr.2026 – Sep.2026

MiniMax Intern

- Developed agent framework for vulnerability discovery and exploitation, integrating headless decompilation with IDA and enhanced GDB-based debugging, achieving a **71.3%** success rate on CyberGym.
- Discovered **9 previously unknown** and **100+** 1-day vulnerabilities across mainstream IoT devices (e.g., Tenda, TOTOLINK, and TP-Link).

Skills

- **Programming:** Python, C, JAVA, Git
- **Security Tools:** gdb, IDA, Binwalk, QEMU
- **Language:** English (IELTS 6.5)
- **Vulnerabilities**: Submitting **62 Independent CVEs**
- **Machine Learning:** PyTorch, HuggingFace Transformers (fine-tuning, RL-based training)
- **Certifications:** CISP-PTE (Certified Information Security Professional – Penetration Testing Engineer), China Information Technology Security Evaluation Center
- **Interests:** Team Sports like Basketball and Soccer.